

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

19ECE34 – EMBEDDED LINUX

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

- | | | | | |
|--|--------------|---------------|------------|------------|
| 1. a) What are the functionalities of cat command in Unix? Given an example to demonstrate each functionality.
b) Illustrate the functioning of grep command in Unix? write a shell script to search a pattern in a file. Pass the pattern to be searched as an input argument. List the pattern matching with the file content. Prompt message to the user if no pattern found or no argument is passed. | 10

10 | L*2

L3 | 1

2 | 1

1 |
| 2. a) How exist status of any Unix command is verified? Illustrate the usage of exist status with an example script
b) Write a shell script to read user's name and year of birth. Display his/her age on standard output. Store the name, year of birth and age in the file. Pass the file name as an input parameter. Display the error message if file already exist. | 10

10 | L3

L3 | 2

2 | 1

1 |
| 3. a) Write a Unix shell script to demonstrate the following programming features with an example
i) case statement ii) if statement
b) How unix will handle positional parameters? Illustrate the way of handling positional parameters with example script. | 10

10 | L3

L3 | 2

2 | 1

1 |

Unit – II

- | | | | | |
|--|--------------|--------------|------------|------------|
| 4. a) How Linux kernel is organized in Linux OS distribution? Explain the functionality of each block.
b) Write a C program and use system call to create a file. Write "welcome to embedded linux" on to it. Pass the file name as the input argument. Display error message if file already exist. | 10

10 | L2

L3 | 3

4 | 1

1 |
| 5. a) With an example, illustrate the following Unix system calls
i) open ii) write
b) Develop a Linux kernel module which display "hello world". Explain the procedure to load and remove developed kernel module | 10

10 | L3

L3 | 4

3 | 1

1 |
| 6. a) Write a C program using system calls to display the file size on the standard output. Pass the file name as an argument. Display the error message if file doesn't exist.
b) What is Linux kernel module? What are the procedure involved in developing Linux kernel module? Explain the procedure to load and removing the Linux kernel Module | 10

10 | L3

L3 | 3

4 | 1

1 |

P.T.O.

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Unit – I

	Marks	BT*	CO*	PO*
1. a) What are the functionalities of cat command in Unix? Given an example to demonstrate each functionality.	10	L*2	1	1
b) Illustrate the functioning of grep command in Unix? write a shell script to search a pattern in a file. Pass the pattern to be searched as an input argument. List the pattern matching with the file content. Prompt message to the user if no pattern found or no argument is passed.	10	L3	2	1
2. a) How exist status of any Unix command is verified? Illustrate the usage of exist status with an example script	10	L3	2	1
b) Write a shell script to read user's name and year of birth. Display his/her age on standard output. Store the name, year of birth and age in the file. Pass the file name as an input parameter. Display the error message if file already exist.	10	L3	2	1
3. a) Write a Unix shell script to demonstrate the following programming features with an example i) case statement ii) if statement	10	L3	2	1
b) How unix will handle positional parameters? Illustrate the way of handling positional parameters with example script.	10	L3	2	1

Unit – II

4. a) How Linux kernel is organized in Linux OS distribution? Explain the functionality of each block.	10	L2	3	1
b) Write a C program and use system call to create a file. Write "welcome to embedded linux" on to it. Pass the file name as the input argument. Display error message if file already exist.	10	L3	4	1
5. a) With an example, illustrate the following Unix system calls i) open ii) write	10	L3	4	1
b) Develop a Linux kernel module which display "hello world". Explain the procedure to load and remove developed kernel module	10	L3	3	1
6. a) Write a C program using system calls to display the file size on the standard output. Pass the file name as an argument. Display the error message if file doesn't exist.	10	L3	3	1
b) What is Linux kernel module? What are the procedure involved in developing Linux kernel module? Explain the procedure to load and removing the Linux kernel Module	10	L3	4	1

P.T.O.

Unit – III

- | | | | | | |
|-------|--|----|----|---|---|
| 7. a) | Illustrate the problem solving 5 steps in embedded software development | 10 | L2 | 5 | 1 |
| b) | Explain the V cycle model in system design with its function block diagram | 10 | L2 | 5 | 1 |
| 8. a) | Explain the procedure involved in adopting water fall model in system design | 10 | L2 | 5 | 1 |
| b) | Distinguish between spiral and V cycle model of system development | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
